Examiner reports

Q1.

The majority recalled that integration was required in part (a), although some differentiated and a few used $s = vt$. A large proportion either forgot the constant of integration or did not calculate its value, losing half the marks.

In part (b), the time of 2 seconds ensured that the toy was stationary when it launched the ball. Many, however, tried to bring extraneous distances and velocities into this question. Making the model fit the situation is an important skill to be assessed. For those who did this, the actual calculation was usually carried out correctly.

Q2.

In part (a) there was a significant proportion of students who scored zero marks and, in many cases, this was because they incorrectly used constant acceleration equations. Most students who realised the problem involved variable acceleration and integration completed the question successfully. For these students the most frequent error was a failure to consider the constant of integration, which generally only cost them one mark.

In part (b), it was surprising to see several students who had used constant acceleration equations in part (a) correctly deciding to integrate in part (b). Students who went on to make a correct comparison with their answer from part (a) could be awarded full marks. Consequently, part (b) had a higher proportion of solutions awarded full marks.

Part (c) was poorly answered. Many students gave unacceptable, non-specific reasons about one minibus taking longer to “get going” or “start-up” than the other. Instead, students had to explain that the times for each bus were so close together that driver reaction time could have a significant effect on the outcome of the test.

Q5.

This question was also answered well by virtually all students who showed that they were familiar with the techniques involved in answering this question. In part (b)(ii) students sometimes failed to find that the value of $-16\pi\sin \pi\, \mathbf{i}$ was zero; this was usually due to students using the angle in degrees rather than radians. The magnitude of $\mathbf{F}$ was often given as –216 rather than +216. In part (c) the majority of students integrated to find the position vector but a common error was to use the vector $4\mathbf{i} - 2\mathbf{j}$ as the + c term. A considerable number of students found the +c term correctly but simplified their final $\mathbf{j}$ term to $-(3t^3 + 79)\,\mathbf{j}$ rather than $+(79 - 3t^3)\,\mathbf{j}$.

Q6.

Nearly all students answered this question well.

Q7.

This question was also answered well by the vast majority of students, who showed that they knew the techniques involved in this question. In part (a) the majority of students integrated to find the velocity of the particle but a common error was to forget the + $\mathbf{c}$ term and thus to ignore the initial position vector. Some found the + $\mathbf{c}$ incorrectly, assuming that

$\mathbf{c}$ was $6\mathbf{i} - 5e^{-4}\,\mathbf{j}$ without completing the necessary calculation.
Another common error was seen whereby students did not integrate the e^{-4t} term correctly.
In part (b) a number of students found the initial velocity, $-15\mathbf{i} - 5\mathbf{j}$, but did not find the initial speed.

Q8.

Most students answered part (a) correctly but some did not replace their a by $\frac{dv}{dt}$. In part (b) students were expected to use $\int \frac{1}{100-v}\,dv = -\int \frac{1}{40}\,dt$; some incorrectly split $\int \frac{1}{100-v}\,dv$ into $\int \frac{1}{100} - \frac{1}{v}\,dv$, whilst others incorrectly used $\int (100 - v)dv = \int \frac{1}{40}\,dt$.

A common error was in forgetting the minus sign in the evaluation of $\int \frac{1}{100-v}\,dv$. Most students attempted to find the '+ c' term in the solution of the differential equation.

Q9.

This question was also answered well. In part (a), a few candidates did not divide **F** by 50 accurately. In part (b), a number of candidates did not find their '+$\mathbf{c}$' correctly, simply replacing their '$\mathbf{c}$' with $7\mathbf{i} - 4\mathbf{j}$. Others, on evaluating their '$\mathbf{c}$', took $-e^{-2t}$ when $t = 0$ to be +1 rather than –1.

Q10.

This question was answered well by most candidates. However, in part (a), a few candidates did not show the required step $0.4\frac{dv}{dt} = 2 - 4v$; others lost and recreated the minus signs, eg $\frac{dv}{dt} = 5 - 10v = -10(0.5 + v) = -10(v - 0.5)$, which was penalised. In part (b), the majority of candidates used $\int \frac{1}{v-0.5}\,dv = -\int 10dt$ and then solved this by integrating both sides.

The integration $\int \frac{1}{v-0.5}\,dv$ caused some candidates difficulty, finding terms such as $\frac{1}{\frac{1}{2}v^2 - 0.5v}$ etc. The use of logarithms was often incorrect, with $\ln(v - 0.5) = -10t + \ln 0.5$ commonly becoming $v - 0.5 = e - 10t + 0.5$

Q11.

This question was also answered well by virtually all candidates, who showed that they knew the techniques involved in answering it. In part (c) the majority of students integrated to find the position vector but a common error was to forget the + c term and thus to ignore the initial position vector. Some found the + c incorrectly assuming that $\mathbf{c}$ was zero without completing the necessary calculation.

Q12.

Most candidates answered part (a) correctly but some just wrote down the printed result.

In part (a) some candidates started with $m\frac{dv}{dt}$ $dv = -9.8v$ but found that this did not give the printed result, and just invented a term in g. This was not accepted. Some were penalised for ignoring the m term or for changing signs.

In part (b) candidates were expected to use $\int \frac{1}{v-5}\,dv = -1.96dt$; some split $\int \frac{1}{v-5}\,dv$ into $\int \frac{1}{v} - \frac{1}{5}\,dv$. However there were many good answers to this part with both the integrations required being successfully completed. The difficulty which some candidates had was in the simplification needed to convert $\ln(v - 5) = -1.96t + \ln 2$ into $v - 5 = 2e^{-1.96t}$ which often became $v - 5 = e^{-1.96t} + 2$

Q13.

Most candidates answered this question well. In part (a), a small number of candidates forgot to add, and find, the +**c** term and thus ignored the initial position vector; a few candidates tried to use equations for constant acceleration. In part (c), some candidates made errors when finding the force by finding $m\mathbf{a}$, which was $2(6t\mathbf{i} - 8\mathbf{j})$, to be $12t\mathbf{i} - 8\mathbf{j}$.

Q14.

Most candidates answered part (a)(i) correctly but some just wrote down the printed result.
In part (a)(ii), candidates needed to start from $F = ma$ and show the cancellation of the mass (65). Some were penalised for ignoring the m term or ignoring the minus sign.

In part (b), candidates were asked to show that $\int \frac{1}{v-2.45}\,dv = -\int 4\,dt$ many simply used this equation and were penalised. There were, however, many good answers to this part, with both the integrations required being successfully completed. The difficulty which candidates had was in the simplification needed to convert $\ln(v - 2.45) = -4t + \ln 17.15$ into
$v - 2.45 = 17.15e^{-4t}$, which often became $v - 2.45 = 17.15 + e^{-4t}$.

Q15.

Most candidates showed that they knew the techniques involved in answering this question. Candidates generally answered part (a) correctly. Part (b)(i) was answered well, but in part (b)(ii) some candidates forgot to find the magnitude of **F**, giving instead just the vector **F**, $-40\mathbf{i} + 30\mathbf{j}$. Most candidates answered part (c) correctly. In part (d) the majority of candidates integrated to find the position vector, but a common error was to forget the +**c** term and thus to ignore the initial position vector. Some found the + **c** incorrectly, assuming that **c** was the initial position vector.

Q16.

This question was answered well; a few candidates had difficulty in the differentiation of $3\cos 4t$.

Q17.

Virtually all candidates answered part (a) correctly. The common problem in part (b) was in the integration of $400\cos\frac{\pi}{2}t$. Apart from this, answers to part (b) were good, and most correctly found t = 2 in part (c). The majority of candidates also found correctly that the speed of the particle was 3, with only a few finishing with the velocity of the particle being –3 or –3**i**.

Q18.

The majority of candidates used $\frac{dv}{dt} = -\frac{\lambda}{v^{\frac{1}{4}}}$ and then solved this by using $\int v^{\frac{1}{4}} dv = \int -\lambda dt$. Some used a = $\frac{dv}{dt}$, together with $v = u + at$, so that $v = u - \frac{\lambda}{v^{\frac{1}{4}}}t$ was often seen.

Many candidates found $\frac{4}{5}v^{\frac{5}{4}} = -\lambda t + \frac{4}{5}y^{\frac{5}{4}}$, but the solution for v was not often found correctly.

A common result was "$v^{\frac{5}{4}} = u^{\frac{5}{4}} - \frac{5}{4}\lambda t \Rightarrow v = u - \left(\frac{5}{4}\lambda t\right)^{\frac{4}{5}}$".

Q19.

This question was usually answered correctly, although some obtained –4cos2t rather than +4cos2t. A common error was the omission of +c, while others added +c but automatically assumed that c = 0.

Q20.

Most of this question was answered well. In part (a), the only difficulty was found in differentiating $2e^{\frac{1}{2}t}$. In part (b), many found the velocity of the particle to be $(e^{\frac{3}{2}} - 8)\mathbf{i}$ [or –3.52**i**], but did not find the speed of +3.52 m s^{-1}. Others gave the speed as 3.5 m s^{-1}, which was penalised. Those candidates who were successful in part (a) usually completed parts (c) and (d) correctly.

Q21.

Virtually all candidates answered this question well, with only a few forgetting to find the magnitude of the force in part (c).

Q22.

Part (a) was usually answered correctly, although some obtained +6cos3t, rather than

$-6\cos 3t$. In part (b), the common error was the omission of $+c$, while others added $+c$ and automatically assumed that $c = 0$.

Q23.

This question was generally answered well. The only common error was in part (b)(ii), where a number of candidates found **F** to be 36**i** + 6**j**, but did not find its magnitude.

Q24.

This question was answered well.

Q25.

Most candidates answered this question well, with only a few making algebraic or arithmetical errors.

Q26.

While candidates realised that part (a) required differentiation, a significant number were not able to carry this out accurately. Part (b) proved more successful, but only a small minority of candidates were able to attempt part (c), the simplest solution involving maximisation of the appropriate trigonometrical function.

Q27.

Most candidates completed this question successfully. A few used $\mathbf{v} = \mathbf{u} + \mathbf{a}t$ in part (b) and obtained $\mathbf{v} = (6 + 3t)\mathbf{i} + (30 - 6t^2)\,\mathbf{j}$, rather than the given answer. After they had done the integration of **a**, a number of candidates 'invented' the 6**i** + 30**j** terms required.

Q28.

Quite a few candidates had problems dealing with the differentiation and integration of the exponential function in this question. They used techniques that were confused and were unable to obtain the correct results. This was probably due to the fact that this question demanded the use of techniques from the core mathematics modules, with which the candidates were not yet fully confident. In part (a)(ii), very few candidates gave fully correct answers for the range of values of the acceleration. Some obtained either the value 2 or the value 14. Of those that did obtain both values, very few used the correct inequalities. In part (b), there were problems with the integration of the exponential function and also with candidates omitting or not find the constant of integration.

Q29.

In part (a), many of the candidates found the force correctly, but some did not go on to give the magnitude. A few candidates did not substitute $t = 0$. In part (b), the main problem was with the integration of the trigonometric terms. A particular problem was with the signs of the functions. Most candidates did attempt to find the constants of integration. A small number of candidates tried to use $\mathbf{v} = \mathbf{u} + \mathbf{a}t$.

Q30.

There were very many good solutions to this question and a good number of candidates scored full marks. Some candidates did however make errors. In part (a) there were some cases where the candidates made some minor errors when differentiating one or both of the components. In part (b) a few candidates had difficulties substituting t = 1/3 correctly, but the biggest problem in this part of the question was the description of the direction. While a number did state that the direction was south, some stated that it was north, even though they had calculated the velocity correctly. Others did not use either north or south, but gave answers that implied that **j** was a vertical unit vector.

Parts (c) and (d) were generally done well, but a few candidates experienced difficulties with the differentiation and very occasionally with the substitution of t = 4 .

Q31.

The candidates generally found part (a) of this question very difficult. Some of the more able candidates found an expression for F in terms of t and then applied Newton's second law. A few effectively reversed this process, by changing the graph into one for acceleration and then finding the equation of the line. Due to the fact that the answer was given, there were a large number of partial or very unconvincing arguments. Some candidates simply omitted this part of the question and went straight on to part (b).

There were many good responses to part (b), but a common error was to ignore the constant of integration. Candidates do need to include a constant of integration and find its value, even if it is zero, in order to gain full marks.

There was a similar problem in part (c), with many candidates again ignoring the constant of integration. Those candidates who included limits of integration, experienced less difficulty, because they wrote down the values of 0 and 200 and substituted both of them.

A few candidates used constant acceleration equations, but the number doing this was fairly small.

Part (d) produced some very mixed responses. There were some very good explanations, but some were very confused. Some candidates wrote about the acceleration instead of the velocity. In some cases the candidates seemed to have some idea of how to respond, but did not express this clearly. Quite a few candidates said that the constant would change, which seemed to imply that they were thinking about the intercept of the graph, but this did not lead to any marks as the velocity does not contain a constant term.

Q32.

Part (a) was frequently done well, with the most common error in part (a)(ii) being the introduction into the equation of motion of an additional force, usually the weight of the motorcycle and rider. Part (b) seldom produced assumptions backed by sound reasons. The majority of responses referred to information given in the question. In part (c), the most convincing solutions were based on the algebraic link between the force and the radius. There was a tendency in part (c) to assume that the magnitude of the frictional force could be altered by changing the radius of the circle, rather than the necessity to adjust the radius to compensate for the force.

Q33.

Many candidates proved competent in handling vector expressions, but a significant number lacked appreciation of vector and scalar quantities and the necessary notation.

Errors in differentiation of trigonometrical functions, particularly in signs, also spoiled otherwise excellent solutions. Otherwise part (a) was done well, with the most frequent error being in the failure to present an answer in radians in part (a)(iii). Responses in part (b)(i) were variable, with scalar answers frequently presented, while in part (b)(ii), few appreciated the variable nature of the direction of the force, and many seeming unaware of both upper and lower bounds of the magnitude of the force.

Q34.

For those who were well prepared in the calculus techniques linking the quantities involved, this question proved highly successful. Part (c)(ii) proved discriminating, as expected, with only the most able candidates linking the components of the velocity vector with the given vector in an appropriate way.

Q35.

Most could attempt differentiation correctly to find an expression for the velocity in part (a).

However poor algebra and lack of appreciation of the form of scalar quantities let down many in part (b). Correct answers in part (b) were often then reduced to ‘$6 + 2t$’ in part (c) and few completed the request correctly.